PROJECT BREAKDOWN: Portfolio Optimization

Overview

The following table outlines the architectural and functional components of the algorithmic portfolio optimization system, structured to match the design specifications of the Stock Screener project.

Parameter	Description
Goal	Develop an algorithmic portfolio optimization engine to construct risk-adjusted stock portfolios from the Nifty 100 index.
Why Optimization	Enables data-driven asset allocation, balancing expected returns with volatility using quantitative models rather than intuition.
Tech Stack	**Python** (core logic), **PyPortfolioOpt** (financial optimization), **Pandas & NumPy** (data manipulation), and **Seaborn & Matplotlib** (visualization).
Database & Storage	Utilizes **In-memory Pandas DataFrames** to handle complex, dynamic time-series financial data efficiently during runtime.
Data Source	Live index constituent and sector scraping from **NSE (niftyindices.com)**, paired with historical price data via **Yahoo Finance (`yfinance`)**.
Data Pipeline	Automated extraction, strict data cleaning (dropping assets with >10% missing data), benchmark (Nifty 50) alignment, and returns calculation.
Risk Management	Employs **Ledoit-Wolf Covariance Shrinkage**, strict weight bounds (max 10% per stock, max 30% per sector), and

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	Hierarchical Risk Parity (HRP) to mitigate overexposure.
Current Features	Markowitz Mean-Variance Optimization (Constrained Max Sharpe), HRP clustering, correlation heatmaps, and a backtesting engine computing CAGR, Max Drawdown, and Sharpe Ratio.
Future Scope (V2)	Integration of advanced ML frameworks for return prediction, live trading API connections, and automated dynamic rebalancing.
Timeline	Iterative pipeline development finalized as a comprehensive, end-to-end quantitative notebook.